Project 21 — Flight Delay Prediction via Gradient Boosting and Temporal Deep Learning

Applied Machine Learning Phase

Evaluation — Logistic Regression & XGBoost Baseline

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Anticipating system disruptions 24 hours before scheduled departure.

The Predictive Challenge: T-Minus 24 Hours



The Operational Stakes

Predictive Target: Identifying flights that will arrive more than 15 minutes late.

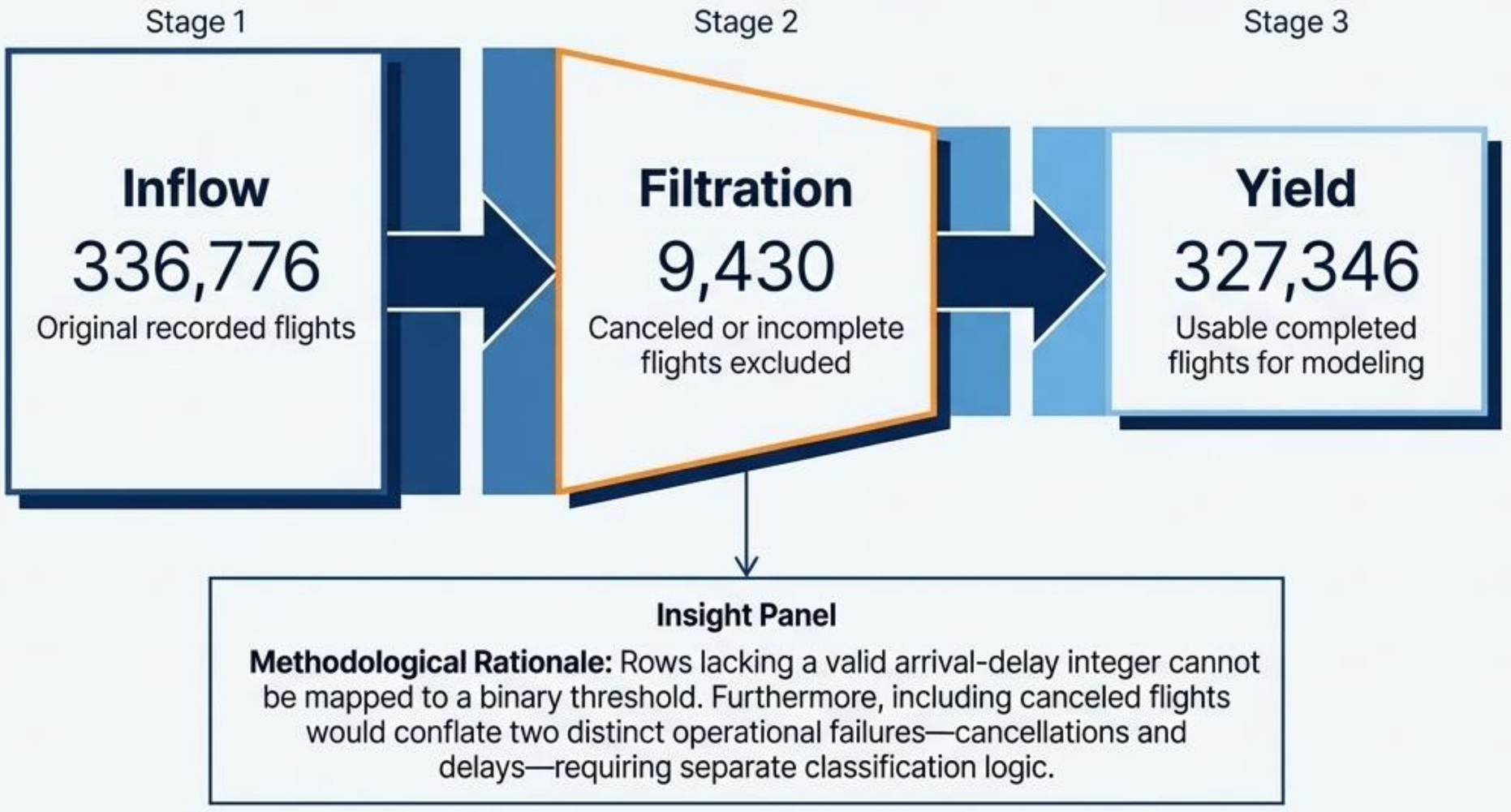
Consequences of Failure: Unanticipated delays cause cascading network failures, sudden gate congestion, and missed passenger connections.

The Predictive Constraint

At 24 hours prior to departure, major influential factors—such as spontaneous weather shifts, day-of mechanical failures, and live airspace congestion—are structurally unknown to the model.

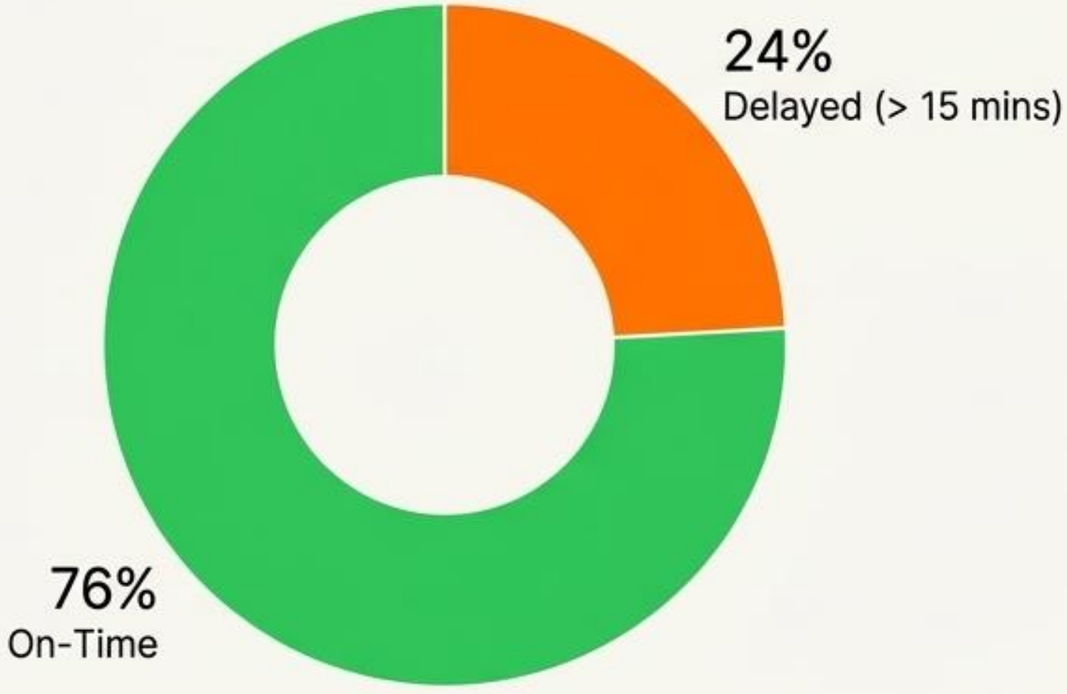
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Isolating valid targets from 2013 US flight records



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The Accuracy Trap: Why predicting ‘on time’ guarantees 76% accuracy but zero operational value.



The Imbalance Reality

Only ~24% of the 327,346 usable completed flights resulted in an arrival delay. **The vast majority of flights operate normally.**

The Deception of Accuracy

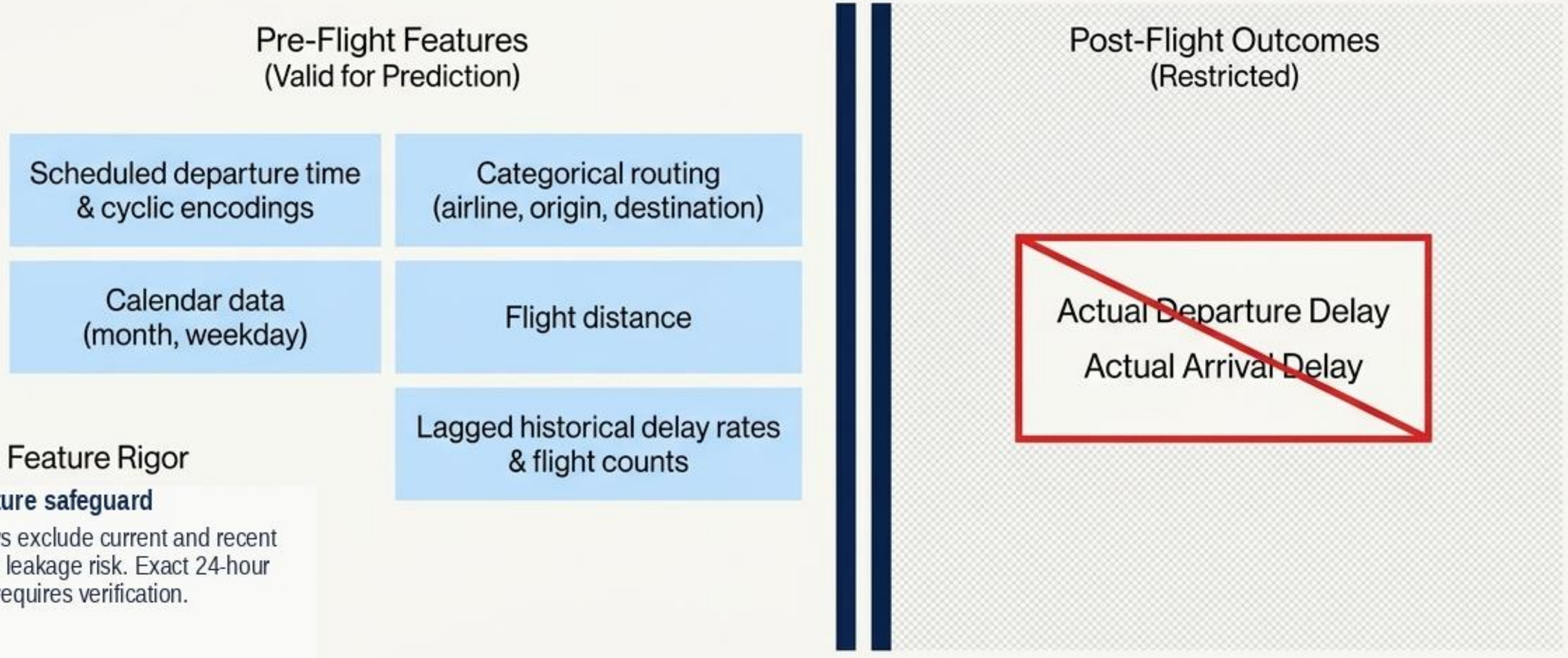
A naive baseline model that hardcodes an 'On-Time' prediction for every single flight will artificially achieve ~76% accuracy, **yet it will miss 100% of the actual disruptions.**

This severe imbalance necessitates evaluating models purely on their ability to detect the minority class, elevating Precision, Recall, PR-AUC, and ROC-AUC as our primary mathematical compass.

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Engineering predictive signals while enforcing strict data leakage prevention

Temporal Barrier Architecture



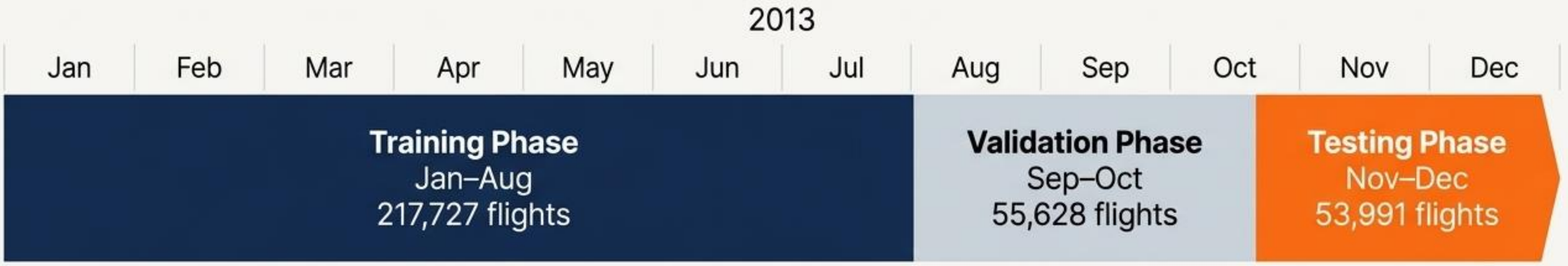
Historical Feature Rigor

Historical feature safeguard

Lagged windows exclude current and recent dates to reduce leakage risk. Exact 24-hour availability still requires verification.

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Forecasting the future requires chronological data splitting, not random sampling.



Methodological Imperative

Randomly shuffling data leaks future seasonal trends (such as December holiday congestion) into the past training set. By splitting chronologically, we force the model to genuinely forecast the 'unknown' future, simulating real-world operational deployment.

Strict Data Hygiene: All preprocessing transformations (scalers, encodings) were mathematically fitted on the Jan–Aug Training data only.

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Methodological framework: Establishing a linear baseline versus a non-linear ensemble

Attribute	Logistic Regression	eXtreme Gradient Boosting (XGBoost)
Mathematical Approach	Linear model of log-odds; sigmoid converts to probability	Ensembled decision trees via gradient boosting
Handling Non-linearities	Requires explicit manual feature engineering	Native capability to map complex variable interactions
Interpretability	High transparent weight mapping	Moderate; requires feature importance extraction
Role in Project Phase 1	The operational baseline	The primary algorithmic challenger

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Translating mathematical evaluation metrics into physical airport operations.

FINAL XGBOOST | OPERATIONAL DECISION THRESHOLD

	Predicted: Not delayed	Predicted: Delayed
	TRUE NEGATIVE **38,049** Correctly identified on-time flights	FALSE POSITIVE **2,450** False delay alerts
	FALSE NEGATIVE **11,853** Actual delays missed	TRUE POSITIVE **1,639** Correct delay alerts

40.08% precision = 40 correct per 100 alerts | 12.15% recall = 12 detected per 100 delays

Baseline vs. Challenger: Performance on unseen chronological test data.

VERIFIED TEST RESULTS — NO ARBITRARY PERFORMANCE THRESHOLDS



XGBoost achieves a modest improvement in delay-risk ranking, not a substantial lift. Its ROC–AUC is 0.6392 and PR–AUC is 0.3553.

Controlling overfitting through targeted hyperparameter optimization.

SELECTED XGBOOST CONFIGURATION

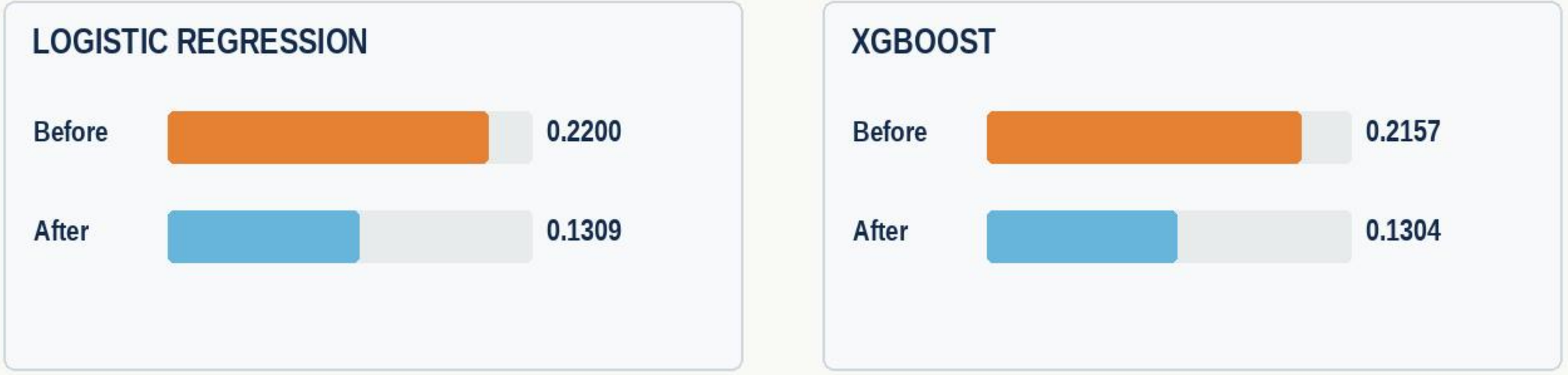
Max depth	4	Learning rate	0.03
Minimum child weight	15	Subsample	0.90
Column subsample	0.90	Best iteration	101

TEST PR–AUC | Earlier XGBoost: 0.3430 → Tuned XGBoost: 0.3553

A modest improvement was observed; reduced overfitting was not independently established.

Probability Calibration: Ensuring the model's confidence matches operational reality.

VALIDATION BRIER SCORE — LOWER IS BETTER



Calibration reduced reported validation-period probability error. It did not directly improve ranking ability. These are validation scores, not final test Brier scores.

Features used for pre-departure prediction

FEATURE GROUPS USED — NO UNSUPPORTED IMPORTANCE RANKING

01 Historical delay rates & flight counts

Lagged origin, airline, destination and route statistics

02 Scheduled departure time

Cyclic representation of departure time

03 Airline & route identifiers

Carrier, origin, destination and route

04 Calendar & flight attributes

Month, weekday and distance

Feature importance requires verified model attribution scores; no causal ranking is claimed here.

The predictive ceiling: Acknowledging the unseen variables of aviation logistics.

Meteorological Volatility

Micro-weather systems and sudden storms developing hours before takeoff.



Mechanical Spontaneity

Unscheduled, day-of aircraft maintenance and sudden equipment failures.

ATC Interventions

Spontaneous ground stops imposed by the FAA due to live, unpredictable airspace congestion.

Methodological Honesty: No model predicting 24 hours in advance can achieve perfect recall. The objective is to map systemic risk, acknowledging that day-of chaos remains an unquantifiable variable in this phase.

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What the final AML models actually achieved

Final Task 7 • 53,991 unseen test flights • Operational precision constraint

TEST METRIC	LOGISTIC REGRESSION	XGBOOST
Accuracy	73.86%	**73.51%**
Precision	41.21%	**40.08%**
Recall	10.78%	**12.15%**
F1-score	17.09%	**18.65%**
ROC-AUC	0.6187	**0.6392**
PR-AUC	0.3387	**0.3553**
Brier score ↓	0.1942	**0.1919**

ACCURACY IS NOT ENOUGH

73.51%

An always-on-time baseline gets ~75.01% on this test set.

PRECISION: ALERT QUALITY

40.08%

About 40 in 100 predicted delay alerts are correct.

RECALL: MISSED DELAYS

12.15%

Only about 12 in 100 actual delayed flights are caught.

CONCLUSION · XGBoost ranks delay risk modestly better, but low recall limits operational usefulness.

Phase II Roadmap: Transitioning to Temporal Deep Learning

FUTURE SCOPE — NOT COMPLETED IN THE AML PHASE

01 REQUIRED DL MODEL
Compact MLP with learned embeddings for airline, origin and destination.

02 OPTIONAL TEMPORAL MODEL
Sequence models only if flight histories are correctly ordered and leakage-safe.

03 BETTER INPUT SIGNALS
Add advance weather forecasts and operational data available at prediction time.

04 WHAT-IF RESCHEDULING
Offline rule-based risk scenarios with explicit assumptions; not yet implemented.